

TASK-1:

```
vennela_07@VennelaPC:~$ cd
vennela_07@VennelaPC:~$ mkdir practical2
vennela_07@VennelaPC:~$ cd practical2
vennela_07@VennelaPC:~/practical2$ mkdir Task1
vennela_07@VennelaPC:~/practical2$ pwd
/home/vennela_07/practical2
vennela_07@VennelaPC:~/practical2$ cd Task1
vennela_07@VennelaPC:~/practical2/Task1$ pwd
/home/vennela_07/practical2/Task1
vennela_07@VennelaPC:~/practical2/Task1$ nano command_executor.c
vennela_07@VennelaPC:~/practical2/Task1$ gcc command_executor.c -o command_executor
vennela_07@VennelaPC:~/practical2/Task1$ ./command_executor
Enter a Linux command: pwd
```

```
Parent Process
Parent PID : 566
Child PID  : 567
```

```
Child Process
Parent PID : 566
Child PID  : 567
```

```
Executing command: pwd
/home/vennela_07/practical2/Task1
```

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if(pid < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }

    else if(pid == 0)
    {
        printf("\nChild Process\n");
        printf("Parent PID : %d\n", getppid());
        printf("Child PID  : %d\n", getpid());

        printf("\nExecuting command: %s\n", command);
    }
}
```

```

else if(pid == 0)
{
    printf("\nChild Process\n");
    printf("Parent PID : %d\n", getppid());
    printf("Child PID  : %d\n", getpid());

    printf("\nExecuting command: %s\n", command);

    execlp(command, command, NULL);

    perror("Execution failed");
    exit(1);
}

else
{
    printf("\nParent Process\n");
    printf("Parent PID : %d\n", getpid());
    printf("Child PID  : %d\n", pid);

    wait(NULL);

    printf("\nChild process completed.\n");
}

return 0;
}

```

TASK-2:

```

vennela_07@VennelaPC:~/practical2/Task1$ cd
vennela_07@VennelaPC:~$ mkdir Task2
vennela_07@VennelaPC:~$ cd Task2
vennela_07@VennelaPC:~/Task2$ uname -a
Linux VennelaPC 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
vennela_07@VennelaPC:~/Task2$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                42 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                       8
On-line CPU(s) list:         0-7
Vendor ID:                   GenuineIntel
Model name:                   Intel(R) Core(TM) Ultra 7 256V
CPU family:                   6
Model:                       189
Thread(s) per core:          1
Core(s) per socket:          8
Socket(s):                    1
Stepping:                     1
BogoMIPS:                     6604.82
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht
                             syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_kno
                             wn_freq pni pclmulqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes x
                             save avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enhanced tpr_shadow
                             ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha
                             _ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir6
                             4b fsrm md_clear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization:               VT-x
Hypervisor vendor:            Microsoft
Virtualization type:          full
Caches (sum of all):
L1d:                           384 KiB (8 instances)
L1i:                           512 KiB (8 instances)
L2:                             20 MiB (8 instances)
L3:                             12 MiB (1 instance)
NUMA:
NUMA node(s):                  1
NUMA node0 CPU(s):            0-7
Vulnerabilities:
Gather data sampling:          Not affected
Ghostwrite:                    Not affected
Indirect target selection:     Not affected
Itlb multihit:                 Not affected

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Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Not affected
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBR SB-eIBRS SW sequence; BHI BHI_DIS_S
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected
vennela_07@VennelaPC:~/Task2$ ps
  PID TTY          TIME CMD
   481 pts/0    00:00:00 bash
   857 pts/0    00:00:00 ps
vennela_07@VennelaPC:~/Task2$ top
top - 03:12:54 up 1:02, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 22 total, 1 running, 21 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7767.6 total, 7112.5 free, 530.8 used, 274.5 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7236.8 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0 21524 12952 9632  S   0.0   0.2   0:01.22 systemd
    2 root        20   0 3180   2208 2072  S   0.0   0.0   0:00.03 init-systemd(Ub
    6 root        20   0 3196   2128 2008  S   0.0   0.0   0:00.00 init
   52 root       19  -1 66840 15948 14760  S   0.0   0.2   0:00.82 systemd-journal
  100 root       20   0 25748  7096 5140  S   0.0   0.1   0:01.72 systemd-udevd
  111 systemd+   20   0 21336 13124 11000  S   0.0   0.2   0:00.12 systemd-resolve
  125 systemd+  20   0 91028  7992  7024  S   0.0   0.1   0:00.37 systemd-timesyn
  168 root       20   0  4244   2684 2432  S   0.0   0.0   0:00.02 cron
  169 message+  20   0  9636   5372 4664  S   0.0   0.1   0:00.26 dbus-daemon
  188 root       20   0 17968  8848  7808  S   0.0   0.1   0:00.12 systemd-logind
  196 root       20   0 1756104 11804 9272  S   0.0   0.1   0:00.50 wsl-pro-service
  209 syslog    20   0 222516  6064 4644  S   0.0   0.1   0:00.14 rsyslogd
  219 root       20   0  3124   1976 1836  S   0.0   0.0   0:00.00 agetty
  235 root       20   0 107024 23140 13676  S   0.0   0.3   0:00.10 unattended-upgr
  300 root       20   0  6704   4672 3884  S   0.0   0.1   0:00.01 login
  344 vennela+  20   0 20108 11384 9448  S   0.0   0.1   0:00.15 systemd
  345 vennela+  20   0 21156   3576 1852  S   0.0   0.0   0:00.00 (sd-pam)
  374 vennela+  20   0 6080   5340 3676  S   0.0   0.1   0:00.02 bash
  477 root       20   0  3188   1120  980  S   0.0   0.0   0:00.00 SessionLeader
  479 root       20   0  3204   1256 1108  S   0.0   0.0   0:00.13 Relay(481)
  481 vennela+  20   0 6080   5372 3656  S   0.0   0.1   0:00.13 bash
  860 vennela+  20   0  9432   5784 3500  R   0.0   0.1   0:00.25 top
```

Operating System is system software that acts as an interface between the user and the computer hardware. Instead of allowing programs to communicate directly with hardware, the OS provides an abstraction layer that makes hardware resources easier to use, secure, and efficient.

The CPU can execute only one instruction at a time on a single core. However, users can run multiple applications simultaneously. The Operating System creates the illusion that every program has its own CPU by using processes, threads, and CPU scheduling.

Memory abstraction means each program feels as though it owns a large, private memory space. The Operating System manages physical ram and provides Virtual Memory.

The Operating System hides the physical details of disks and provides a File System. Instead of accessing disk sectors directly, users work with:

- Files
- Folders

- Directories

Different hardware devices have different communication methods. The Operating System hides these differences using Device Drivers. Applications use the same system calls regardless of the hardware.